

Step 1: Install Git on Your System (Windows)

`git --version`

Step 2: Set Up Git with Your Name and Email

Run these commands in Git Bash or terminal:

```
git config --global user.name "Your Name"
git config --global user.email your-email@example.com
```

Step 3: Create a Private Repository on GitHub

1. Log in to <https://github.com>
2. Click the + icon in the top-right → click **New repository**
3. Enter a repository name (e.g., `my-private-project`)
4. Select **Private**
5. Do **not** add a README or `.gitignore` (keep it empty)
6. Click **Create repository**

Step 3: Generate SSH Key on Your System

```
ssh-keygen -t rsa -b 4096 -C "your-email@example.com"
```

- Replace `"your-email@example.com"` with your actual GitHub email.
- When it asks:
 - **Enter file to save the key:** Just press **Enter** to save in the default location:
`C:\Users\YourName\.ssh\id_rsa`
 - **Enter passphrase:** You can leave it empty (just press Enter twice), or enter a secure passphrase.

This will generate:

- `id_rsa` (your private key)
- `id_rsa.pub` (your public key)

Step 4: Copy the SSH Public Key

To copy the SSH key, use:

```
clip < ~/.ssh/id_rsa.pub
```

This will copy the key to your clipboard.

If `clip` doesn't work, you can open the file manually:

```
cat ~/.ssh/id_rsa.pub
```

Then copy the output manually.

Step 5: Add SSH Key to GitHub

1. Go to <https://github.com>
2. Login → Click your profile picture → **Settings**
3. Left menu → **SSH and GPG keys**
4. Click **New SSH key**
5. **Title:** Give any name like `My Laptop Key`
6. **Key:** Paste the copied key (`id_rsa.pub` content)
7. Click **Add SSH key**

Step 6: Test SSH Connection

In Git Bash, run:

```
ssh -T git@github.com
```

If everything is set up correctly, you will see:

```
Hi your-username! You've successfully authenticated, but
GitHub does not provide shell access.
```

Step 5: Use SSH URL to Push Code (Open git in same dir)

```
git add .
git commit -m "Initial commit"
git remote add origin https://github.com/your-username/my-
private-project.git
git push -u origin master
```

When asked for a username and password:

- Enter your GitHub username
- For password, paste the **Personal Access Token**

Basic Git Commands

Command	Description
<code>git init</code>	Initialize a new Git repository in the current folder
<code>git clone <repo-url></code>	Clone an existing repository to your system
<code>git status</code>	Show the current state of the working directory and staging area

Command	Description
<code>git add <file></code>	Add a specific file to the staging area
<code>git add .</code>	Add all files (new/changed) to the staging area
<code>git commit -m "message"</code>	Commit staged changes with a message
<code>git log</code>	View commit history
<code>git diff</code>	Show changes between working directory and staging area

Working with Remote Repositories

Command	Description
<code>git remote -v</code>	Show remote URLs connected to the repo
<code>git remote add origin <repo-url></code>	Add a remote repository
<code>git push -u origin master</code>	Push commits to the remote <code>master</code> branch (first time)
<code>git push</code>	Push latest changes to remote
<code>git pull</code>	Fetch and merge changes from remote to local
<code>git fetch</code>	Download changes from remote but don't merge

Branching and Merging

Command	Description
<code>git branch</code>	List all branches
<code>git branch <branch-name></code>	Create a new branch
<code>git checkout <branch-name></code>	Switch to another branch
<code>git checkout -b <branch-name></code>	Create and switch to new branch

Command	Description
<code>git merge <branch-name></code>	Merge a branch into the current branch
<code>git branch -d <branch-name></code>	Delete a branch (locally)

Undo / Reset Changes

Command	Description
<code>git reset <file></code>	Unstage a file (keep changes in working directory)
<code>git reset --hard</code>	Discard all uncommitted changes
<code>git restore <file></code>	Discard changes in working directory for a file
<code>git checkout <commit-id></code>	Switch to a previous commit (detached head state)

Viewing and Managing Files

Command	Description
<code>git ls-files</code>	List all tracked files
<code>git rm <file></code>	Remove a file from working directory and staging area
<code>git mv old new</code>	Rename or move a file in Git
<code>git config --list</code>	Show all Git configurations